



iConspicuity & ADS-L Update

Aircraft Electronics Association – Europe
Regional Meeting
13 November 2025

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iConspicuity Project Manager

Established
2002



EASA

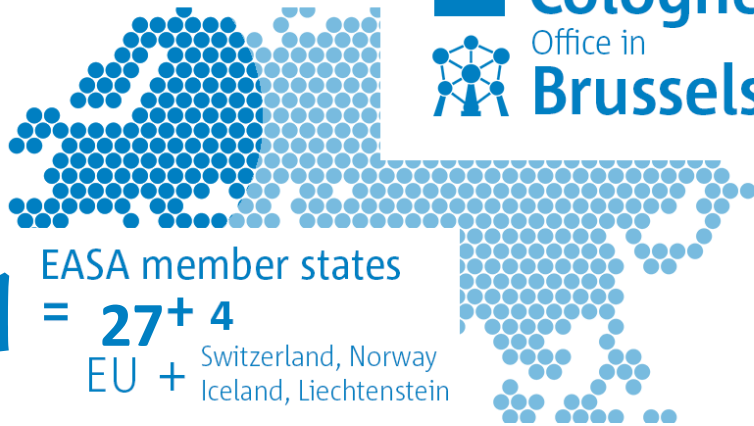
European Union Aviation Safety Agency

800+

aviation experts
& administrators



Headquarters in
Cologne
Office in
Brussels



31 EASA member states
= 27+ 4
EU + Switzerland, Norway
Iceland, Liechtenstein



GA Flightpath 2030+



Safety

Enhancing safety culture

Sustainability

Enabling sustainable growth

Digitalization

Embracing a digital future to maximize the benefits of technology

Inclusiveness

Encouraging wider participation and accessibility

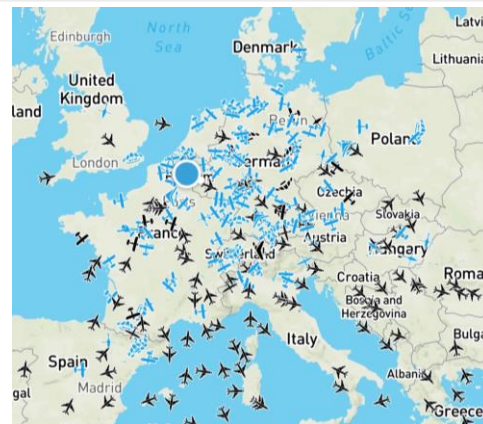


Greener Faster

EASA Policy

Stimulate availability of:

- Fast-charging infrastructure
- Sustainable / Synthetic Fuels



Fly Direct

EASA Guidelines

- Implementation of Aeroplane PinS (proceed VFR)
- Harmonizing measures for protecting species and habitats from aviation

Declared by Default

EASA Policy

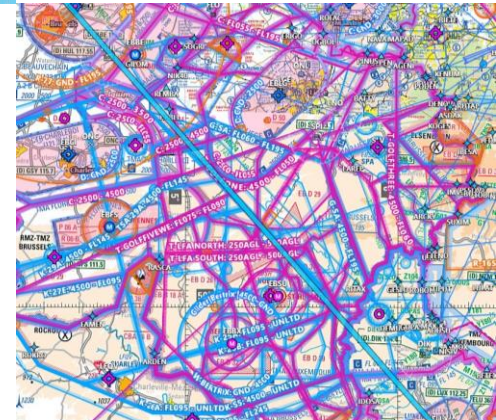
Declaratory system for sport, recreational and private aviation activities based on industry standards rather than prescriptive rules and prior certification or approval



iConspicuity a Reality

Joint Roadmap with EUROCONTROL

- Updated ADS-L Specification
- iConspicuity ATM Research
- ADS-L *Coalition*
- iConspicuity Declaration



Strategy 2025-2028



“Your safety is our mission”

SAFE



Maintain a safe,
resilient aviation
ecosystem

GLOBAL



Enhance global
standards and
interoperability

GREEN



Achieve net zero
emissions in
aviation by 2050

INNOVATIVE



Integrate
innovative
technologies

RESILIENT



Equip EASA for
the next decade



Safety Risk Management

Efficient SRM process that develops mitigations for the top safety issues through EPAS (focus on safety).



Risk Based Oversight

DOA mechanism designed & endorsed. Principles for other organisations harmonised. Maturity model for NCAs.



ATM & Aerodromes

Certification and oversight of ATM DPO, ATM/ADR ground equipment.



Implement SMS & ISMS across Industry

Full integration into organisation oversight & standardisation.
Competence building for NCA/EASA.



Flightpath 2030+

Deliver Programme Objectives related to e-conspicuity and prevention of mid-air collision.



Maintain a safe, resilient aviation ecosystem

Ensure that European safety and oversight standards effectively mitigate all risks affecting safety.

Approve safe products that perform as expected in the system as part of a competitive industry.

Promote a human-centered approach to maintaining safety.

iConspicuity



JOINT HIGH-LEVEL ROADMAP



Use Cases

Reduce collisions

and other airborne hazards by
enhancing situational awareness of
their surroundings



Access U-space

through **affordable and interoperable electronic conspicuity** for manned aircraft operating without ATC services



Additional benefits*

Complement FIS and SAR without
requiring changes to existing
ATM/ANS principles and/or
operational practices



Target Solution



Simple

System design that ensures interoperability and affordability for end users



One Language

To ensure interoperability.
ADS-B and ADS-L are good candidates for common language(s)



One Link

Air-Ground transmission for U-space
A direct radio Air-Air link for pilot awareness
A second link for other purposes

Candidate Technologies



1090 MHz

Worldwide used protected spectrum requiring certification and licensing



978 MHz

Regionally deployed protected spectrum requiring certification and licensing. **Needs evaluation and planning in Europe**



SRD 860

Unprotected and unlicensed but **regulated and standardised** spectrum **currently available** in all Europe until ~2030



Mobile Telephony

Possible complementary link in altitudes where network is available, but not supporting direct air-to-air and **requiring network**

Approach



Consider

Key criteria

U-space **mandate**

Voluntary elsewhere

Different needs (IFR vs glider)

Dual use cases (e.g. ADS-B for ATC and U-space)



Communicate

Throughout the process

A **clear strategy and communication** campaign to get stakeholders to implement the right solutions

Address

Use Cases

Pilots' **situational awareness**,
Europe-wide at all altitudes

U-space conspicuity, initially
geographically limited & low
altitude



Assess

Candidate technologies

'One link' based on a
comparison of options
considering **assessment of**
ground-based **operations**
and the business case for all
users (airborne and on the
ground)



Timeline & Implementation Milestones





Before 2020

Non-interoperable or private solutions available for GA, no solutions for drones, iConspicuity concept developed by EASA

2020-2023

EASA teamed up with industry in development of ADS-L as a solution for GA in U-space airspace

2024-2027

Interoperable (ADS-L) protocol developed tested for air-to-air, U-space and selected ATM use cases. Required spectrum needs finalised.

2028 onwards

Any pilot or operator can choose an interoperable and affordable solution to benefit from enhanced situational awareness in any airspace.

2025 Top Priorities



Communication & Promotion

Roadmap: Phase A-E

EPAS: SPT.119 SPT.0120 MST.0027

2025 Deliverables

- ✓ Sunny Swift - ADS-L: see and be seen
- ✓ iConspicuity webpage
- ✓ iConspicuity Declaration
 - For EASA, NAAs, GA Associations
- ✓ ADS-L Coalition
 - For Industry



iConspicuity Declaration

Introduction

This iConspicuity Declaration is a voluntary policy statement, jointly established by participating aviation authorities and other entities, specifically addressed to the General Aviation (GA) sector. Its primary objective is to facilitate and encourage the use of iConspicuity devices and data (including ADS-B out, ADS-C, surveillance data, and similar information) by all stakeholders with the shared aim of improving operational safety and enhancing safety culture in GA.

The adoption of this iConspicuity Declaration does not affect the application of Regulation (EU) No 923/2012 regarding the reporting, analysis and follow-up of occurrences in civil aviation and any other applicable European Union or national legislation.

Commitment

By adopting this iConspicuity Declaration we, the signatories, express our commitment to foster the development of iConspicuity devices and their use and utilization of related data with the intention of improving aviation safety, fostering innovation and enhancing operational efficiency through collaborative analysis. We pledge to adhere to the following:

Key Principles

- 1. Promoting Safety Culture:** The initiative aims to facilitate and promote safety culture in GA in order to foster positive safety behaviours.
- 2. Voluntary nature:** The initiative is a partnership that signatories join on a voluntary basis.
- 3. System-wide insights:** Analysis of iConspicuity data will focus on system-wide insights (big data approach) rather than the actions of specific situations. This encourages broad participation and fosters a safe aviation environment.
- 4. 360-Degree Collaboration:** All relevant stakeholders will be involved in the analysis of iConspicuity data, ensuring a holistic and trusted approach to safety and operational improvements¹. The collaborative analysis will lead to jointly agreed actions that will benefit all participants.
- 5. Transparent Monitoring:** The process for analysing and acting upon iConspicuity data must be transparent, allowing all stakeholders to track progress and ensure alignment with safety improvement goals.
- 6. Data protection:** The use of data and information derived from the initiative will comply with the EU's General Data Protection Regulation² (GDPR), which governs how the personal data of individuals in the EU can be processed and transferred.

Footnotes

- ¹ iConspicuity Definition
- ² Regulation (EU) 2016/679 (GDPR) states that the processing of personal data must be lawful, fair, and transparent to the data subject. It is a principle that the data subject should be informed of the purpose, scope, and other data to other aircraft and ground stations, ensuring operational safety and safety. ADS-B comes in two main types: ADS-B In and ADS-B Out.
- ³ ADS-B is a higher version of ADS-B In message content, uniquely designed to be used as an aircraft's electronic means of communication to other aircraft.
- ⁴ iConspicuity is a term used to describe the use of ADS-B Out data for system-wide insights.
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¹ iConspicuity [website](#)

² Associations representing airspace users and relevant industry

³ ADS-B stands for Automatic Dependent Surveillance-Broadcast. It's a technology that enables aircraft to broadcast their position, speed, and other data to other aircraft and ground stations, enhancing situational awareness and safety. ADS-B comes in two main types: ADS-B Out and ADS-B In.

⁴ ADS-L is a lighter version (subset) of the ADS-B message content, originally developed to make manned aircraft electronically conspicuous when operating in U-space airspace.

⁵ European Plan for Aviation Safety, Volume II, 2024 Edition, task MST.0027 - Promotion of safety culture in GA

⁶ Manned aircraft are not required to be electronically conspicuous, except when operating as uncontrolled traffic in U-space airspace

⁷ This can serve as preparation for future inclusion in the [DataSafety](#) programme.

⁸ Regulation (EU) 2016/679 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data.





SKYRECON



Authorities already supporting the Declaration



EASA



Czech Republic



France



Greece



Latvia



Lithuania



Netherlands



Republic of North Macedonia



Romania



Slovenia



Spain



Switzerland

WP1 – Communication & Promotion

Roadmap: Phase A-E

EPAS: SPT.119 SPT.0120 MST.0027

→ EASA

→ [FAQs Automatic Dependent Surveillance — Light \(ADS-L\)](#)

→ FOCA

→ [Technology recommendation](#)

The proposed open standards are ADS-B out on 1090 MHz and ADS-L in the SRD 860 MHz frequency band. While the former is almost exclusively used in certified installations, the latter can be used in installed or portable devices.

→ [FAQs by FOCA](#)

iConspicuity in ATM (Research)

Roadmap: Phase A

EPAS: RES.0032

2025-2026 Deliverables (with Eurocontrol)

- No change of existing rules, procedures or practices
- ATM use cases & related information elements
- Areas to be further developed/clarified by EASA
- Activities
 - Successful kick-off workshop on 25-Feb (100+ participants)
 - 2+ use case dedicated workshops (FIS, SAR, TMZ/RMZ, aerodrome safety)
 - FIS workshop 2-3 July (Brussels), SAR workshop on 21 November (ICAO Paris)
 - Final dissemination workshop in Q2 2026

WP3 – ADS-L 4 SRD860

Roadmap: Phase B

EPAS: RMT.0690, RMT.0230, RMT.0519

2025 Deliverables

- ✓ Draft Issue 2 - Review and comments consolidation
- ✓ Focus on: U-space, FIS-B, TIS-B, Remote ID (SORA)
- ✓ Comments review meetings (industry leads)
- Publication in Q4 2025
- *Issue 3 will focus on air-to-air!*

Work in Progress

ADS-L

4 SRD860

Issue 2*

Aviation

Traffic

Status

Uplink

Traffic

FIS-B

Drones

RemoteID

NEW

NEW

WP4 – ADS-L 4 MOBILE

Roadmap: Phase B

EPAS: RMT.0690, RMT.0230, RMT.0519

2025 Deliverables

- ✓ LTE Task Force established
- Work on ADS-L 4 MOBILE is about to start

Advice from Sunny Swift

A FRIEND WANTS TO INSTALL SUCH A SYSTEM IN HIS PRIVATE AIRCRAFT. HE'S NOT SURE WHICH SYSTEM TO CHOOSE.

I SUGGEST BUYING AN ADS-L SYSTEM WITH DIRECT RADIO LINE OF SIGHT CAPABILITY, COMPLEMENTED BY MOBILE NETWORK CONNECTIVITY.



Implementations

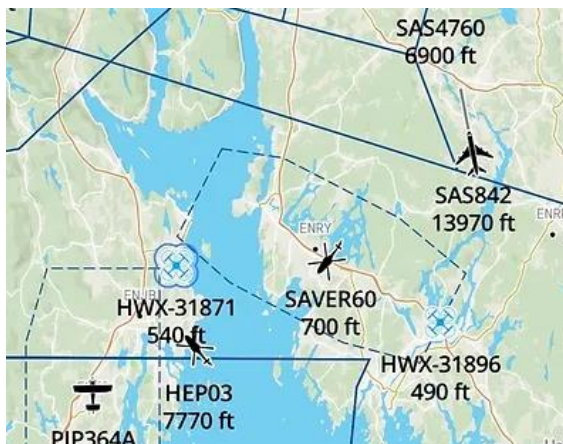
South-Eastern Finland

Two ground stations for reception of position data from various systems (ADS-B, ADS-L, UAT, MLAT, FLARM, RemoteID) and UAT retransmissions of nearby traffic, weather, NOTAMS to GA aircraft



Norway (Oslo)

Five transceivers (ADS-B, ADS-L, UAT, MLAT, FLARM, RemoteID) allow drone pilots to receive alerts from nearby GA aircraft (including helicopters and paragliders) and vice versa



France (La Réunion)

The network of eight transceivers (ADS-B / ADS-L / FLARM / OGN / RemoteID) has doubled the number of conspicuous aircraft in the mountains and in the vicinity of airports

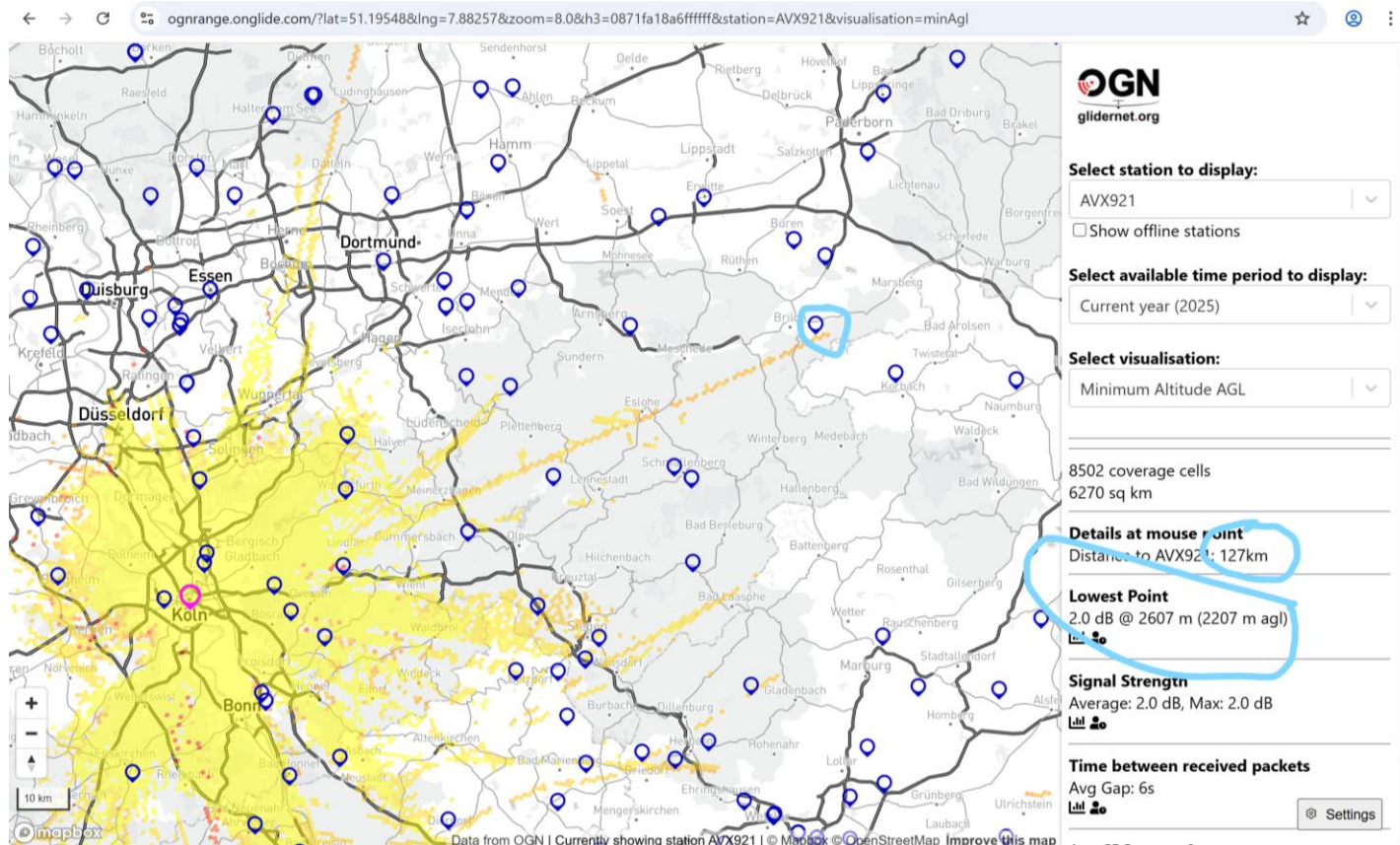


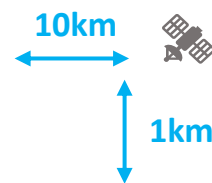
EASA Station

ADS-B, ADS-L, FLARM, PilotAware, OGN, Remote ID, Basic Mode Multilateration



EASA Station - Performance





ADS-L for General Aviation and Drones



IFR

FL195 / 5.900m

VFR + IFR

